A Review Gate Stops Wrong Memory Writes but Not Missed Updates: Candidate Selection in User-Controlled Long-Term Memory for LLM Assistants

Radha Mohan Pattanayak
Associate Professor, SCOPE
VIT-AP University
Amaravati, India
radhamohan.pattanayak@vitap.ac.in

Soubhagya Swastik Dhal
B.Tech (Computer Science & Engineering)
VIT-AP University
Amaravati, India
soubhagya3696@gmail.com

Basant Kumar
B.Tech (Computer Science & Engineering)
VIT-AP University
Amaravati, India
gbasant201036@gmail.com

Mehal Mathur
B.Tech (Computer Science & Engineering)
VIT-AP University
Amaravati, India
mehalmathur24@gmail.com

Aditya Kumar Jena
B.Tech (Computer Science & Engineering)
VIT-AP University
Amaravati, India
adi.jena1904@gmail.com

***Abstract*—Assistants that keep long-term memory about their user increasingly ask for approval before they change what they believe. We ask what such an approval gate does, and does not, protect. We study a local assistant (AURA) in which a language model proposes add, update, retire and forget operations, changes to existing memories wait for the user, and old values are kept as history. In simulation (60 persona runs per cell) the gate removes 94–98% of the true facts that automatic application destroys when the extractor errs, at 20–33 review prompts per 100 turns. It does not reduce stale facts: the extractor can only change memories that are shown in its prompt, and with default candidate selection the memory to be changed is shown for only 0.690 of change requests when 350 unrelated memories are stored, and for 0.286 when the change shares a message with other sentences. A candidate selection that searches sentence by sentence without a minimum score raises the latter to 0.866. On two real Gemini models (12 personas each) the change target was missing from the prompt in 25 and 22 of 136 change turns; with the new selection it was never missing, stale facts per persona fell from about 1 to 0 under the gate (sign test $p = 0.002$ and 0.002), and F1 rose by +0.077 and +0.057, at about 35% more prompt tokens and no significant increase in review prompts. A rule that lets a rejected change return after the user says something new gave no measurable benefit with real models. On 30 LongMemEval knowledge-update questions the gain with the gate (+0.167) is not significant ($p = 0.063$), and the memory system stays below full-context answering. The evaluation harness and all result files are provided.**

***Index Terms*—long-term memory, conversational assistants, large language models, human-in-the-loop, memory update, evaluation.**

I. Introduction

Large language model (LLM) assistants that remember facts about their user across conversations must decide, after every message, whether to add a fact, change one they already hold, or drop one. Mistakes are costly in two different ways. A *wrong write* replaces or removes something that was true ("moved to Pune" overwrites the true employer), and a *missed update* leaves an outdated value active after the user has said that it changed. A growing line of work treats wrong writes as a governance problem and proposes checks before a memory is stored [1]. A direct form of such a check is to ask the user: changes to existing memories become proposals that the user accepts or rejects, and old values are kept as history.

This paper starts from a simple observation about where such a gate sits. It acts on the operations that the extractor model *returns*. An update that the extractor never produces cannot be proposed, and therefore cannot be approved. In a retrieval-based design the extractor can change only the memories that are placed in its prompt, so the choice of these candidates bounds how many updates are possible at all. We therefore ask three questions about a user-controlled memory: (Q1) How much of the damage of extractor errors does the gate remove, and at what cost in user prompts? (Q2) How often is the memory that a change refers to visible to the extractor, and how does this depend on the size of the store and on how the candidates are chosen? (Q3) Does a better candidate selection translate into fewer stale facts with real models, and what does it cost?

We study these questions in AURA, a local assistant with a SQLite memory store, local sentence embeddings and a Gemini extractor, using the assistant's real server code throughout. Our contributions are:

1) We separate wrong writes from missed updates and measure both under a controlled error model. The gate cuts destroyed true facts by 94–98% but leaves stale facts

essentially unchanged (Sec. V).
2) We measure *reachability*, the share of change requests whose target memory is in the extractor's prompt, over store sizes from 0 to 350 and for single-sentence and mixed messages, and show that the default selection becomes the bottleneck (Sec. VII).
3) We describe two small switches, a candidate selection that searches per sentence without a minimum score (F2) and a bounded re-proposal rule (F1), and evaluate them in an ablation in simulation and with two real Gemini models, including their cost. F2 carries the effect; F1 does not show a measurable benefit with real models (Sec. VIII–IX).
4) We report an external check on LongMemEval knowledge-update questions and state clearly what it does and does not show (Sec. X), and we provide the harness, the personas generator and all raw results.

A concrete case from our data (Sec. IX-B) shows the problem. A persona says "I drive a Seltos" in one session and, three sessions later, "I sold my old car and bought a Swift." With default settings the extractor was shown neither the stored "Seltos" memory (the sentence is a poor match for it) nor a reason to look for it; it added "bought a Swift" as a new fact, and the outdated Seltos memory stayed active. The review gate never saw an update to approve. The same persona under the new selection produced the update, the user approved it, and the history was kept.

The rest of the paper is organised as follows. Section II places the work among related studies, Section III describes the memory system, and Section IV the evaluation. Sections V–X give results in the order policies, retrieval, reachability, ablation, real extractors and LongMemEval; Sections XI–XIII discuss implications, limits and future work.

We deliberately keep claims narrow. The personas are synthetic and English, the real-model samples are small, the user is simulated, and no user study was run (Sec. XII).

II. Related Work

Memory for conversational agents and its benchmarks. Mem0 describes an architecture that extracts, consolidates and retrieves salient facts from conversations and evaluates it on LoCoMo [2], a benchmark of very long agent–agent conversations with question answering, summarisation and multimodal tasks [3]. LongMemEval benchmarks chat assistants on five long-term memory abilities, among them reasoning about updated facts, and splits memory design into indexing, retrieval and reading [4]. Our work uses LongMemEval's knowledge-update questions as an external check; the benchmark does not label which proposed memory changes are right, which is why our gate there uses an oracle approximation (Sec. X).

Updating and governing memory. Lam et al. propose a governance framework (SSGM) that separates memory updates from task execution and applies consistency checks and permission controls before anything is stored [1]. Patel studies the *memory-update gap*, the loss of accuracy on questions about facts that changed, and addresses it by reinforcement-learning fine-tuning of a small model [5]. Xiong et al. show that adding and deleting stored experiences changes agent behaviour and that errors can compound through retrieval [6], and Xiao et al. find that errors entering through memory updates persist longer than errors entering through user questions [7]. Zhang et al. place a learned gate between memory store and model that decides which *retrieved* memories are admitted [8]. Our setting differs in three ways: the gate here is a user decision on *writes*, the problem we isolate is upstream of the gate (what the extractor can see), and the remedy is inference-time candidate selection rather than training. Chu surveys security threats to agents across layers and notes that current benchmarks rarely test failures that span sessions [9].

TABLE I
What the cited works study (from their abstracts).

Work	Focus
Mem0 [2]	extraction, consolidation and retrieval of conversation facts; evaluated on LoCoMo
LoCoMo [3]	very long agent conversations; QA, summarisation, multimodal tasks
LongMemEval [4]	five memory abilities, among them updated facts; indexing–retrieval–reading split
SSGM [1]	governance framework: consistency checks and permission controls before storing
Supersede [5]	accuracy loss on changed facts; reinforcement-learning environment
Xiong et al. [6]	effect of adding and deleting experiences; error compounding
Xiao et al. [7]	how memory-update errors propagate over turns
MemGate [8]	learned gate over retrieved memories
Chu [9]	survey of agent security threats by layer

Positioning. Table I summarises what each cited line of work studies, as far as the abstract pages show. Two observations motivate the present study: update-related failures are reported as a problem for agent memory [5], [7], and governance proposals add checks on the write path [1], but a measurement of how a write-time user gate interacts with the *visibility* of the memory to be changed is, to our knowledge, not among them. We cannot exclude that such a measurement exists elsewhere.

Scope. The works above were read at the level of their arXiv abstract pages for this paper, not in full; their detailed claims should be checked before they are relied upon. We make no claim about priority over any of them.

III. System: User-Controlled Memory in AURA

A. Memory store and operations

AURA keeps memories in SQLite. A memory has text, a category (preference, personal, people, work, plan, other), a status (`active`, `superseded` or `outdated`), a pointer to the memory that replaced it, a source and its conversation, and timestamps. After each user message a Gemini model (default `gemini-3.5-flash-lite`) receives the message and a list of candidate memories and returns operations: *add*, *update* (replace a value), *retire* (the fact no longer holds), or *forget*. An update keeps the old row as `superseded` with a link to its successor; retire marks the old row `outdated`. The extractor can refer only to memory identifiers that appear in its prompt.

TABLE II
MEMORY AND PROPOSAL RECORDS.

Field	Meaning
`content`	one third-person sentence, with date context for time-bound facts
`category`	preference, personal, people, work, plan, other
`status`	`active`, `superseded` (replaced), `outdated` (no longer true)
`superseded_by`	identifier of the replacing memory
`source`, conversation	where the fact came from
proposal: action	`update`, `retire` or `forget`
proposal: status	`pending`, `accepted`, `rejected` (with decision time)

B. Data model

Table II lists the fields that matter for the evaluation. Nothing is overwritten in place by an update: the old row stays with status `superseded` and points to its successor, so every change can be audited and, if needed, reversed by the user.

C. Extraction prompt

After each batch of new user messages the extractor receives the date, the candidate memories as lines of the form `[id] (category) text`, optional earlier context and the new messages. The instructions tell it to store only what the *user* said or confirmed (preferences, personal details, people, work, plans, routines), not questions, general knowledge, assistant suggestions or one-off requests, to write one short third-person sentence per fact, and to choose among the four operations. The output is constrained to a JSON schema with the fields action, memory_id, content and category. An operation that names an identifier that was not in the candidate list is ignored by the application. This is the point at which candidate selection limits what can change.

D. The review gate

When "ask before changing memories" is on (the default), *add* operations are applied directly and every operation that touches an existing memory becomes a stored *proposal* (action, target memory, proposed text, conversation). The user accepts or rejects it; nothing changes until then. We call this policy **gate**. The comparison policies are **append** (only additions; a changed fact is stored as a second fact), **auto** (all operations applied at once), and **gate all** (also new facts wait for approval; emulated in the harness, not an AURA setting).

E. Candidate selection

The prompt contains candidate memories chosen from the store. By default these are the 12 best search matches for the user's message plus the 8 most recently created memories. Search combines the cosine similarity of MiniLM sentence embeddings [10] (`all-MiniLM-L6-v2`, run locally) and a lexical overlap score, $0.85 \cdot \cos + 0.15 \cdot \text{lex}$, and drops matches below a minimum score of 0.25. We evaluate four modes, selected by an environment variable:

- **default**: as above;
- **no minimum**: the same, without the 0.25 cut-off;

Algorithm 1: Candidate selection (modes *default* and *adaptive*)

Input: user text x, mode, store, minimum score $\tau = 0.25$
Output: candidate memories shown to the extractor

$f \leftarrow 0$ if mode is no-minimum or adaptive, else τ;
if *mode is sentence or adaptive* **then**
  $k \leftarrow 24$ if mode is adaptive and x contains a change cue, else 12;
  foreach $q \in \{x\} \cup \text{sentences}(x)$ **do**
    foreach *memory* m *in* $\text{search}(q, k, f)$ **do**
      keep the best score per memory;
  $R \leftarrow$ the k memories with the best scores;
else
  $R \leftarrow \text{search}(x, 12, f)$;
return $R \cup$ the 8 most recent memories;

TABLE III
WHAT IS REAL AND WHAT IS SIMULATED.

Part	Sim. runs	Real runs
Store, operations, proposals, search	AURA code	AURA code
Sentence embeddings	MiniLM	MiniLM
Extractor	simulated, rate e	Gemini
Reviewing user	simulated, slip u	simulated, $u = 0$
Conversations	scripted personas	same personas

- **sentence**: the message is split into sentences and each is searched on its own, so that an unrelated sentence does not dilute the query;
- **adaptive** (F2): sentence search without minimum score, and a larger number of matches (up to 24) when the message contains a change cue (words such as "moved", "stopped", "no longer", "forget").

Using F2 enlarges the prompt (Sec. IX); the minimum score exists to keep unrelated memories out (Fig. 3).

F. Re-proposal (F1)

A rejected proposal normally ends the matter. With F1 a rejected change may be proposed again, at most N times, but only after the user has said something new about it (we use $N = 2$). The aim is to recover from a rejection that was a slip without nagging the user. Concretely, when the extractor repeats a rejected change, the application compares the time of the last user message with the time of the last rejection and creates a new proposal only if the user has written since and the number of earlier rejections is at most N. An identical proposal that is still pending is never duplicated, and a newer proposal for the same memory replaces an older pending one.

IV. EVALUATION METHOD

All experiments run AURA's real server code, real SQLite schema and real local embedding model on a throw-away database. What is real and what is simulated is listed in Table III.

A. Personas and ground truth

A persona has 8 of 12 fact types (city, employer, dog, sister, diet, gym time, language, car, project, coffee, hobby, birthday)

TABLE IV
PERSONA 1 OF THE REAL-MODEL RUNS: WHAT THE USER SAYS IN EACH SESSION, AND THE TRUE STATE AT THE END.

- Session 1: 'I go to the gym at 7 am every day.' 'My birthday is on 21 September.' 'I take my coffee as a cappuccino.' 'My dog is called Mithu.'
- Session 2: 'Like I said, my dog is called Mithu.' 'My friend Rohan just bought a Thar, it looks great.' 'I live in Kolkata.' 'I drive a Seltos.' 'I work at Swiggy.' 'I'm leading a project called Lotus at work.'
- Session 3: 'Like I said, i go to the gym at 7 am every day.' 'What's the weather like in Shimla today?' 'I moved my gym time to 6 pm.' 'My project wrapped up last week, I'm between projects.'
- Session 4: 'Like I said, i work at Swiggy.' 'Please forget what I told you about my birthday.' 'Can you set a timer for ten minutes?' 'I've just moved to Pune.'
- Session 5: 'Is it true that octopuses have three hearts?' 'Like I said, i work at Swiggy.' 'I've just moved to Chennai.' 'I sold my old car and bought a Swift.'
- Truth at the end: gym: 6 pm; birthday: no longer true; coffee: as a cappuccino; pet: Mithu; city: Chennai; car: Swift; employer: Swiggy; project: no longer true. Asked to forget: birthday (21 September).

and five sessions of about 23 user turns. Facts are stated, later updated ("I've just moved to Hyderabad"), retired ("I stopped going to the gym"), forgotten on request, repeated, and mixed with distractors that must not become memories (hypotheticals, questions, assistant suggestions, facts about friends). A store starts with 30 unrelated true background facts and gains 10 between sessions, imitating earlier use. The true state is known after every turn. Table IV lists one persona in full; it is used again in the worked example (Sec. IX-B).

In the ablation runs a restating turn ("Like I said, ...") follows each update, which is what F1 needs to act on.

B. Simulated extractor and user

The simulated extractor returns what a correct model would return for the turn, using only the memories shown in the prompt; a change whose target is not shown becomes an add, as a real model would do. Errors are injected at rate e: a correct operation is dropped, a distractor becomes a false memory, or a true memory shown to the model is retired or overwritten with a false value. All policies face the same random draws. The simulated user reviews proposals at the end of a session, accepts correct ones and rejects wrong ones, and errs with probability u (default 0.05).

C. Scoring rules

Memories written by a model are scored by the *values* they mention (for example "Pune", "Swift", "6 am"), using the persona's slots and a list of alternative wordings. A memory mentioning a value that is true at the end is *true*. One that mentions a value the user gave earlier and that is no longer true is *stale*, unless it records that the fact ended ("finished the project and is between projects", after a retire request), which is *ended* and counted as correct. A memory about a distractor value is *hypothetical* when it is worded as a hypothetical, *third-party* when it concerns another person (allowed), and *false* otherwise. A memory that mentions no known value is *other*. The exact lists are given in Appendix A.

D. Metrics

F1 compares the persona's active memories with the true facts. **Stale** counts active memories holding a value the user gave earlier that is no longer true. **False** counts active memories the user never stated about themselves. **Destroyed** counts true memories, persona and background, that ended up superseded, outdated or deleted by a wrong change; F1 does not see the background memories, so this is the safety metric. **Prompts/100** is the number of review cards per 100 user turns. **Reachability** is the share of change requests whose target memory is among the extractor's candidates. **Prompt tokens** are summed over a persona run. With T the set of persona facts that are true at the end, A the set of active persona memories scored true or ended, and P the set of all active persona memories, precision is $|A|/|P|$, recall is the share of T covered, and F1 their harmonic mean. **Exact state** is 1 for a persona whose final memory has every true fact, no stale and no false fact, and 0 otherwise.

E. Real-model runs and scoring

Real runs use Gemini as the extractor for 12 personas, one seed, with the simulated user as an oracle ($u = 0$), for the policies auto and gate and four variants (baseline, F1, F2, F1+F2). They were repeated for baseline and F1+F2 with a second model (`gemini-3.1-flash-lite`). Because models word memories freely, a memory is scored by value: the automatic scorer maps each active memory to a persona fact and a label (true, stale, ended, hypothetical, third party, other). The author hand-labelled every memory of the baseline and F1+F2 runs (195 and 175 memories). There were 3324 extraction calls and 0 failed. Requests were rate-limited (12 per minute), each persona was played in its own isolated database, and the extractor was the only part of the system that called the model; the reviewing user was the oracle described above and judged a proposal by the same value-based rules as the scorer. This has a consequence that we discuss with an example (Sec. IX-B): the oracle can reject a correct proposal whose wording the rules do not recognise.

F. Statistics

Differences between variants are paired by persona and seed. We report the mean difference with a 95% bootstrap interval (4000 resamples, fixed seed) and an exact two-sided sign test on the non-tied pairs. No correction for multiple tests is applied; the tests run are 210 (simulated ablation) and 48 (real runs), so isolated p-values near 0.03 should not be read as findings. Reachability is reported with Wilson intervals [11], and agreement between hand and automatic labels with Cohen's κ [12]; since there is a single annotator this measures the scorer against the annotator, not human–human agreement.

V. Results: What the Gate Protects

Fig. 1 and Table V compare the four policies over extractor error rates e (60 persona runs per cell: 20 personas, 3 seeds).

Wrong writes. Under auto-apply the extractor's errors destroy true facts: 0.78, 1.60, 2.83 and 4.60 per persona at $e = 0.05, 0.1, 0.2, 0.3$. The gate reduces this to 0.05, 0.03, 0.18 and 0.25, i.e. by 94–98%. The remainder comes from the simulated user's slips. The price is 20–33 review prompts per 100 turns; requiring approval for new facts as well removes false new facts but needs about 2.7 times as many prompts as the gate.

Missed updates. The gate does not help here. At $e = 0$ the append-only policy leaves 5.00 stale facts per persona by construction, and the gate still leaves 1.80, slightly more than auto-apply (1.57); across the error sweep the gate leaves between 1.8 and 2.2 stale facts, auto-apply between 1.6 and 1.9. F1 of the gate is below auto-apply at $e = 0$ (0.854 vs. 0.881) and about equal at higher e. The gate thus buys safety at a small cost in F1, and does nothing for stale facts.

Earlier evidence for the candidate bottleneck. An earlier run of the same simulation (without restating turns) already pointed at candidate selection: removing the 0.25 minimum score from the extractor's candidate search, and nothing else, roughly halved the stale facts left at low error rates under both auto-apply and the gate (Fig. 2). That a change in how candidates are chosen moves the stale count far more than the choice between gate and auto-apply is what prompted the reachability analysis below.

User slips. Table VI varies the simulated user's slip rate at $e = 0.1$. F1 falls from 0.808 to 0.732 and stale facts rise from 1.75 to 2.52 as u goes from 0 to 0.3: a gate is only as good as the review.

VI. Results: Does Memory Search Find the Right Memory?

Candidate selection builds on memory search, so we first measured search alone. Each store holds one persona's current facts plus $N \in \{25, 100, 350\}$ unrelated facts; every stored fact is asked about with two natural questions ("Which company do I work for?"), 2178 queries in all over 20 personas and 3 seeds. A query is a hit when the right memory is among the top six results. We compare AURA's hybrid score with the 0.25 minimum, the same score without a minimum, embeddings only and keywords only (the fallback used when the embedding model is unavailable). A check that re-scoring reproduces AURA's real top-six lists succeeded for all queries (differences only from ties).

Table VII shows that AURA's search finds the right fact for only 0.700 of the questions at $N = 350$ and returns nothing at all for 0.281, while the same score without a minimum reaches 0.891. The cause is visible in the scores: for 0.293 of the questions the right fact scores below 0.25 on the hybrid score, because a natural question about a fact ("What do I do for fun?") shares few words and only moderate embedding similarity with the stored sentence ("The user's main hobby is chess").

The failure is concentrated in a few kinds of fact (Table VIII). Facts with a distinctive topic word (dog, sister, birthday, coffee, language) are found almost always; facts that people phrase in many ways or that name no topic word (city, employer, diet, car) are often not found with the 0.25 minimum, although the no-minimum search finds most of them.

The minimum score is there for a reason. Table IX and Fig. 3 show the trade-off: lowering it raises the share of stored facts found but also the share of questions about things the user never said that still return memories (noise), up to 1.00 with no minimum. A single threshold therefore has to compromise between recall and silence, and at 0.25 AURA returns something for about half of the never-stated questions at $N = 350$ while finding only 0.70 of the stored ones. For a conversational recall tool this may be tolerable; for the extractor, whose job is to see the memory it must change, it is a bottleneck.

VII. Results: What the Extractor Can See

For each of 30 stores per size we generated update, retire and forget requests (2610 in all) and checked whether the memory the request refers to is among the extractor's candidates. The store holds N unrelated memories besides the ones the persona has stated.

Reachability falls from 0.898 with an empty store to 0.690 at $N = 350$ (Fig. 4, Table X). Even with an empty store about one change in ten is out of reach because the 0.25 minimum drops weak matches. Without the minimum the figure is 0.911 at $N = 350$. The recent-memory slots help only at small N.

Table XI compares the modes, with the mean number of candidates in parentheses. When the change shares the message with two unrelated sentences, which is common in natural conversation, default selection reaches only 0.286 of changes at $N = 350$, because the embedding of the whole message is dominated by the unrelated content. Removing the minimum score gives 0.607; per-sentence search alone 0.690; and the adaptive mode 0.866, at 28.1 candidates on average instead of 8.3. Per-sentence search makes a mixed message behave like a single sentence. The minimum score is not free to remove, because it also suppresses unrelated matches (Table IX), and a larger prompt costs tokens.

By kind of change. Table XII splits the result at $N = 350$ by update, retire and forget requests. Retire and forget requests are reached more often than updates by the default selection in single-sentence turns, but all three suffer in mixed turns, and the adaptive mode restores them to a similar level. (Single-sentence turns have one sentence, so the sentence mode equals the default there by construction.)

By kind of fact. The aggregate hides large differences (Table XIII, Fig. 5). For single-sentence changes at $N = 350$ the default selection reaches the car in only 0.04 of cases, the employer in 0.18 and the city in 0.34; the adaptive mode raises these to 0.84, 0.95 and 0.49. The car case is the one in our worked example. The city remains the weakest even with F2: a statement such as "We relocated to Pune last month" contains no word that points to the old city, and with 350 other memories the right one is often not among the 24 best matches.

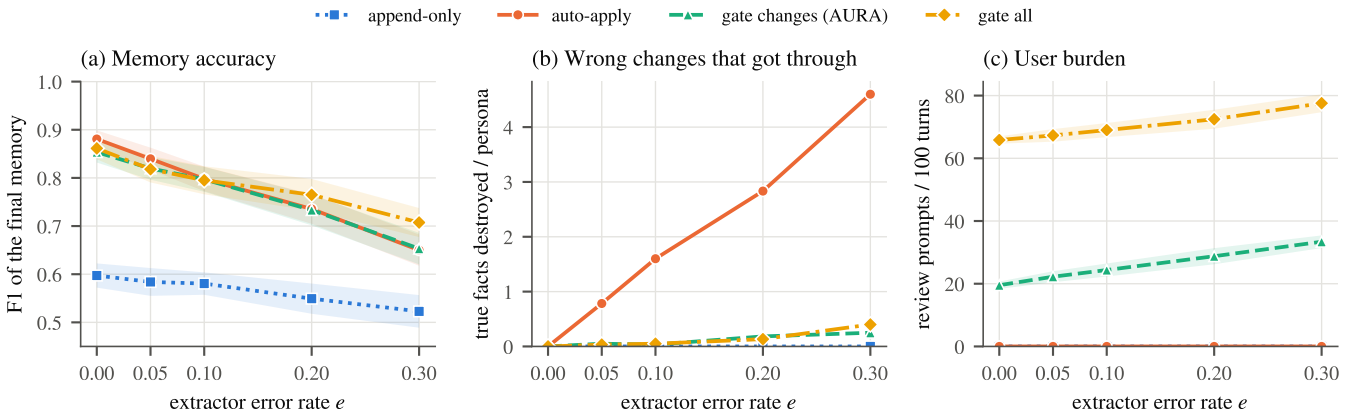


Fig. 1. Policies under extractor errors (simulated, shaded: 95% CI). (a) F1. (b) True facts destroyed by wrong changes. (c) Review prompts per 100 turns.

TABLE V
REVIEW POLICIES VERSUS EXTRACTOR ERROR RATE e (SIMULATED, USER SLIP $u = 0.05$, MEAN PER PERSONA).

e	policy	F1	stale	false	destroyed	prompts/100
0	append-only	0.597 ± 0.025	5.00	0.00	0.00	0.0
	auto-apply	0.881 ± 0.018	1.57	0.00	0.00	0.0
	gate changes	0.854 ± 0.022	1.80	0.00	0.00	19.5
	gate all	0.861 ± 0.023	1.62	0.00	0.00	65.8
0.1	append-only	0.581 ± 0.023	4.58	0.33	0.00	0.0
	auto-apply	0.798 ± 0.026	1.68	0.43	1.60	0.0
	gate changes	0.797 ± 0.026	1.85	0.33	0.03	24.4
	gate all	0.795 ± 0.029	1.83	0.02	0.05	69.0
0.3	append-only	0.523 ± 0.034	3.87	1.20	0.00	0.0
	auto-apply	0.650 ± 0.032	1.92	1.33	4.60	0.0
	gate changes	0.653 ± 0.033	2.18	1.22	0.25	33.4
	gate all	0.707 ± 0.030	1.98	0.07	0.40	77.6

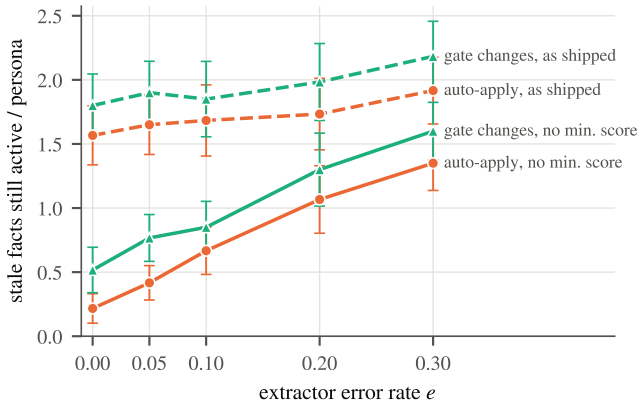


Fig. 2. Stale facts left active, as shipped versus candidate search without the minimum score (earlier simulation run, no restating turns).

F2 is therefore an improvement, not a complete solution, and a store with many place-like facts will need better candidate selection than ours.

TABLE VI
GATE UNDER USER SLIPS (SIMULATED, $e = 0.1$).

slip u	F1	stale	wrong changes applied	prompts/100
0	0.808 ± 0.026	1.75	0.02	24.5
0.1	0.783 ± 0.028	2.00	0.28	24.4
0.2	0.757 ± 0.027	2.33	0.42	24.0
0.3	0.732 ± 0.029	2.52	0.62	24.1

VIII. RESULTS: ABLATION OF THE TWO SWITCHES

A. Simulated

Table XIV and Fig. 6 give the paired ablation under the gate (error sweep, $u = 0.05$, restating turns on). F2 reduces stale facts per persona at $e = 0.1$ from 1.83 to 0.65 and F1+F2 to 0.58; the F2 effects are significant ($p < 0.001$) in every cell shown. F1 alone moves F1 by +0.012 at $e = 0$ ($p = 0.031$), and is not significant at $e = 0.1$ and 0.2. With 210 tests that single p-value would not survive a correction. F2 raises the review load by about 5 prompts per 100 turns at $e = 0.1$ because more changes are discovered and proposed.

IX. RESULTS: REAL GEMINI EXTRACTORS

Table XV and Fig. 7 give the same ablation with a real extractor (12 personas, one seed, oracle user). Table XVI repeats baseline and F1+F2 with a second model.

TABLE VII
RETRIEVAL: SHARE OF QUESTIONS WHOSE ANSWER MEMORY IS IN THE TOP SIX.

N	search variant	hit@6 [95% CI]	top-1	no result
25	AURA (hybrid, min 0.25)	0.707 [0.672, 0.739]	0.705	0.287
	hybrid, no minimum	0.994 [0.986, 0.998]	0.920	0.000
	embedding only, no minimum	0.994 [0.986, 0.998]	0.897	0.000
	keywords only (fallback)	0.606 [0.570, 0.641]	0.581	0.335
100	AURA (hybrid, min 0.25)	0.705 [0.671, 0.737]	0.690	0.281
	hybrid, no minimum	0.968 [0.953, 0.979]	0.861	0.000
	embedding only, no minimum	0.957 [0.940, 0.970]	0.828	0.000
	keywords only (fallback)	0.606 [0.570, 0.641]	0.576	0.325
350	AURA (hybrid, min 0.25)	0.700 [0.665, 0.732]	0.687	0.281
	hybrid, no minimum	0.891 [0.866, 0.912]	0.833	0.000
	embedding only, no minimum	0.872 [0.846, 0.894]	0.787	0.000
	keywords only (fallback)	0.606 [0.570, 0.641]	0.567	0.309

TABLE VIII
RETRIEVAL BY KIND OF FACT AT $N = 350$ (HIT@6; MEAN COSINE SIMILARITY OF THE RIGHT FACT).

fact	queries	AURA hit@6	no-min. hit@6	mean cosine
city	60	0.27	0.37	0.253
employer	76	0.24	0.89	0.248
dog	68	1.00	1.00	0.595
sister	44	1.00	1.00	0.582
diet	56	0.09	0.41	0.175
gym time	68	1.00	1.00	0.480
language	66	1.00	1.00	0.494
car	64	0.34	1.00	0.236
project	62	1.00	1.00	0.436
coffee	56	1.00	1.00	0.483
hobby	66	0.65	1.00	0.400
birthday	40	1.00	1.00	0.602

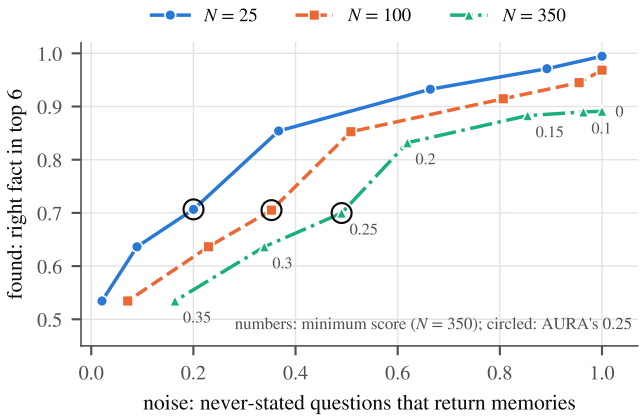


Fig. 3. Retrieval trade-off of the minimum score: share of stored facts found in the top six versus share of never-stated questions that still return memories. Circled: AURA's 0.25.

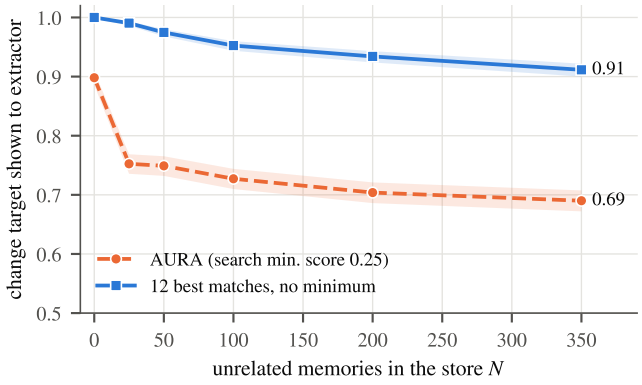


Fig. 4. Share of change requests whose target memory is shown to the extractor, versus store size N (shaded: Wilson 95% CI).

Reachability on real conversations. With the baseline selection, the target memory was missing from the prompt on 25 of 136 change turns (first model; both policies together) and 22 of 136 (second model). With F1+F2 it was missing on 0 and 0. This is the mechanism of Sec. VII observed with real conversations.

Stale facts and F1. Under the gate, stale facts per persona fell from 1.00 to 0.00 (first model; $p = 0.002$) and from 1.08 to 0.00 (second; $p = 0.002$), and F1 rose from 0.905 to 0.982 (+0.077, $p = 0.021$) and from 0.845 to 0.901 (+0.057, $p = 0.002$). F2 alone reached 0.33 stale facts ($p = 0.021$) and F1 of 0.953 ($p = 0.146$, not significant); the benefit of adding F1 to F2 was not tested separately and is within what 12 personas can resolve. F1 alone changed nothing measurable (stale 1.08, $p = 1.000$): on real models F1 is not supported. Under auto-apply the picture is the same (stale 1.00 to 0.00, $p = 0.002$).

Safety on real models. No injected errors exist in real runs, so we count what the models did. Across all auto-apply persona runs (72), 7 true background facts were destroyed by the model's own operations (5 of them in the two baseline runs); across the 72 gate runs the count was 0. With the oracle user, no correct change was rejected, no wrong change was accepted and 0 wrong changes were applied. Because the user is perfect and the counts are small, this is a consistency check on the simulation, not a measurement of real users.

Cost. Prompt tokens per persona rose from about 13.4k to 18.1k (+35%, $p < 0.001$), and the number of prompts per 100 turns from 23.3 to 24.9 (first model, $p = 0.289$) and from 24.5 to 26.0 (second, $p = 0.180$), not significant. Of the change prompts, 68 of 69 (first model) and 68 of 72 (second) in the

TABLE IX
MINIMUM SCORE TRADE-OFF: FOUND / NOISE FOR N AND MINIMUM SCORE. FOUND: SHARE OF STORED-FACT QUESTIONS ANSWERED; NOISE: SHARE OF NEVER-STATED QUESTIONS THAT STILL RETURN MEMORIES.

N	0	0.1	0.15	0.2	0.25	0.3	0.35
25	0.99/1.00	0.97/0.89	0.93/0.66	0.85/0.37	0.71/0.20	0.64/0.09	0.53/0.02
100	0.97/1.00	0.94/0.96	0.91/0.81	0.85/0.51	0.71/0.35	0.64/0.23	0.53/0.07
350	0.89/1.00	0.89/0.96	0.88/0.85	0.83/0.62	0.70/0.49	0.64/0.34	0.53/0.16

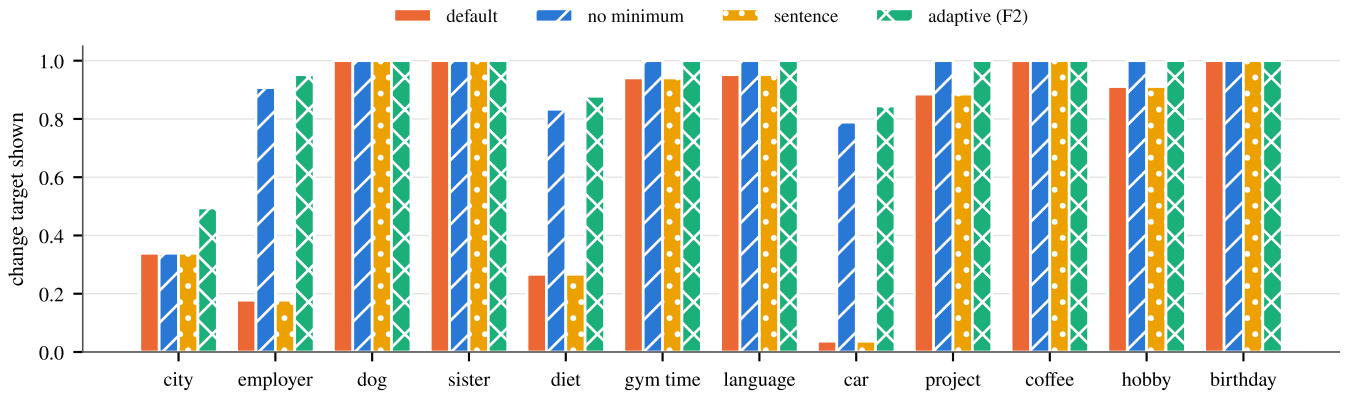


Fig. 5. Share of change requests whose target memory is shown to the extractor, by kind of fact ($N = 350$, single-sentence changes, 4 selection modes).

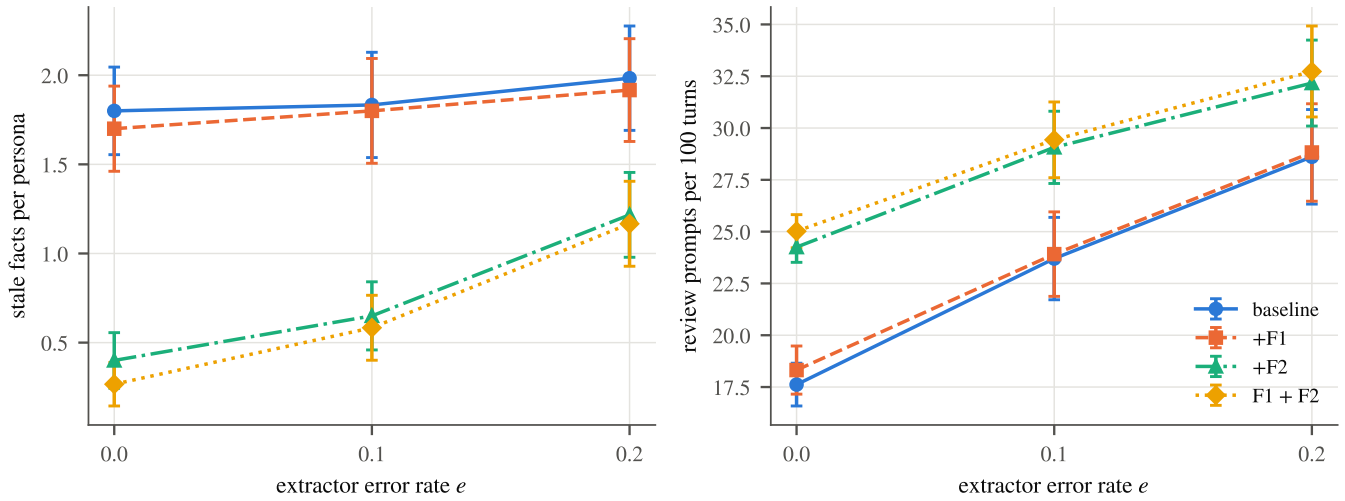


Fig. 6. Gate policy, error sweep: stale facts and review prompts for baseline, F1, F2, F1+F2 (simulated).

F1+F2 runs were correct changes that the user accepted, against 56 of 65 and 55 of 68 for the baselines; this is consistent with the added prompts being updates that the baseline missed.

Other effects. Under auto-apply the F2-only variant stored 3 false facts across 12 personas where the baseline stored none, and F1+F2 stored none again; with these counts we cannot say whether this is an effect of F2 or noise.

A. *Per-persona results*

Table XVII lists every persona of the first model under the gate. With the baseline selection, 10 of 12 personas ended with at least one stale fact and 10 had at least one change target missing from the prompt; with F1+F2 none did. Persona 9 has the same F1 under both settings. Persona 3 has a lower F1 under F1+F2 because the model stored one more hypothetical sentence ("would like to live in Goa someday"), which lowers precision; none of its updates is wrong, so this is not a regression in updating.

B. *A worked example*

Persona 1 (Table IV) is a typical case. Table XVIII shows the proposals made under the baseline, Table XIX the final memories, and Tables XX and XXI the same under F1+F2. Under the baseline the extractor updated the gym time, the

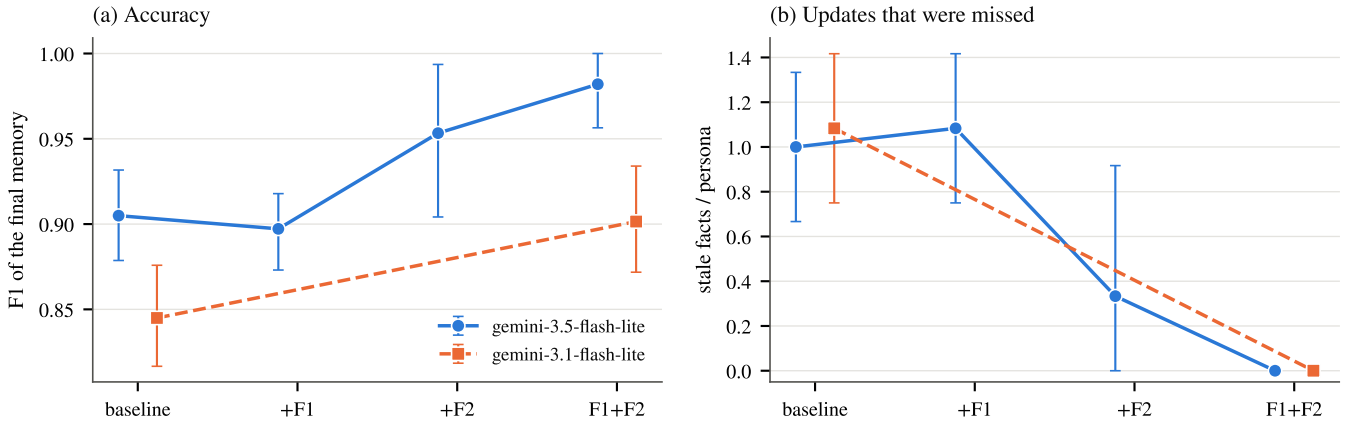


Fig. 7. Gate policy with real Gemini extractors: (a) F1, (b) stale facts per persona; mean and 95% bootstrap interval over 12 personas.

TABLE X
REACHABILITY FOR SINGLE-SENTENCE CHANGES (95% WILSON INTERVAL).

N	shown (default)	shown (no minimum score)
0	0.898 [0.886, 0.909]	1.000 [0.999, 1.000]
25	0.752 [0.736, 0.769]	0.990 [0.986, 0.994]
50	0.749 [0.732, 0.765]	0.975 [0.968, 0.980]
100	0.727 [0.710, 0.744]	0.952 [0.944, 0.960]
200	0.704 [0.686, 0.721]	0.934 [0.924, 0.943]
350	0.690 [0.672, 0.707]	0.911 [0.900, 0.922]

TABLE XI
REACHABILITY BY SELECTION MODE (MEAN NUMBER OF CANDIDATES IN PARENTHESES).

N	turn type	default	no min.	sentence	adaptive
0	single sentence	0.898 (8.2)	1.000 (11.4)	0.898 (8.2)	1.000 (11.4)
	mixed turn	0.747 (8.1)	0.998 (11.4)	0.898 (8.2)	1.000 (11.9)
350	single sentence	0.690 (11.3)	0.911 (19.8)	0.690 (11.3)	0.936 (28.2)
	mixed turn	0.286 (8.3)	0.607 (19.8)	0.690 (11.3)	0.866 (28.1)

city (twice) and forgot the birthday, but it never proposed to replace “drives a Seltos”; the user’s “I sold my old car and bought a Swift” became a separate fact (“bought a Swift car”) and the Seltos memory remained active and stale. The one change target missing from the baseline prompt on this persona is consistent with this. A further proposal, replacing the project memory by “currently between projects”, was rejected by the oracle reviewer. Under F1+F2 the same persona produced the car update, and the project memory was replaced by an “ended” record that was accepted.

The rejected proposal shows a limitation of our reviewer. The oracle judges proposals with the same value-based rules as the scorer, and a correct paraphrase can be rejected when the rules do not recognise its wording. In the baseline run this happened once for this persona; the final memory then still holds the old project memory, which the scorer rightly counts as stale. We therefore treat the oracle as an approximation of a careful user, not as one.

TABLE XII
REACHABILITY BY KIND OF CHANGE AT $N = 350$.

turn type	change	default	no min.	sentence	adaptive
single sentence	update	0.734	0.907	0.734	0.943
	retire	0.697	0.989	0.697	0.995
	forget	0.649	0.870	0.649	0.894
mixed turn	update	0.314	0.610	0.734	0.894
	retire	0.241	0.567	0.697	0.906
	forget	0.288	0.628	0.649	0.819

TABLE XIII
REACHABILITY BY KIND OF FACT AT $N = 350$ (SINGLE-SENTENCE CHANGES).

fact	changes	default	no min.	sentence	adaptive
city	180	0.34	0.34	0.34	0.49
employer	270	0.18	0.91	0.18	0.95
dog	270	1.00	1.00	1.00	1.00
sister	90	1.00	1.00	1.00	1.00
diet	180	0.27	0.83	0.27	0.88
gym time	270	0.94	1.00	0.94	1.00
language	270	0.95	1.00	0.95	1.00
car	270	0.04	0.79	0.04	0.84
project	270	0.89	1.00	0.89	1.00
coffee	270	1.00	1.00	1.00	1.00
hobby	180	0.91	1.00	0.91	1.00
birthday	90	1.00	1.00	1.00	1.00

C. Cost in requests, tokens and time

Table XXII gives the cost per persona. The number of requests is the same (one extraction per user turn); prompt tokens grow by about 35% because more candidates are listed; output tokens are unchanged. Measured time per request is not a reliable cost measure here: it changed from 5.61 s to 9.39 s with the first model but from 6.15 s to 5.74 s with the second, which indicates that API load during the runs, not the prompt size, dominated.

D. Scorer validity

Table XXIII shows the agreement of the automatic labels with the author’s hand labels. Three of 195 + 175 memories

TABLE XIV
ABLATION UNDER THE GATE (SIMULATED). IN PARENTHESES: DIFFERENCE TO THE BASELINE AND SIGN-TEST p (60 PAIRS).

e	variant	F1 (diff., p)	stale (diff., p)	prompts/100
0	baseline	0.854	1.80	17.6
	+F1	0.865 (+0.012, 0.031)	1.70 (−0.10, 0.031)	18.3
	+F2	0.950 (+0.096, <0.001)	0.40 (−1.40, <0.001)	24.3
	F1+F2	0.967 (+0.114, <0.001)	0.27 (−1.53, <0.001)	25.0
0.1	baseline	0.817	1.83	23.7
	+F1	0.821 (+0.005, 0.500)	1.80 (−0.03, 0.500)	23.9
	+F2	0.890 (+0.074, <0.001)	0.65 (−1.18, <0.001)	29.1
	F1+F2	0.901 (+0.084, <0.001)	0.58 (−1.25, <0.001)	29.4
0.2	baseline	0.737	1.98	28.6
	+F1	0.747 (+0.010, 0.250)	1.92 (−0.07, 0.250)	28.8
	+F2	0.776 (+0.039, <0.001)	1.22 (−0.77, <0.001)	32.2
	F1+F2	0.784 (+0.047, <0.001)	1.17 (−0.82, <0.001)	32.7

TABLE XV
REAL EXTRACTOR `GEMINI-3.5-FLASH-LITE`: 12 PAIRED PERSONAS PER CELL. IN PARENTHESES: DIFFERENCE TO THE BASELINE AND EXACT SIGN-TEST p.

policy	variant	F1	stale	false	prompts/100	prompt tokens ($\times 10^3$)
auto	baseline	0.908	1.00	0.00	0.0	13.2
	+F1	0.902 (−0.007, 1.000)	1.00 (+0.00, 1.000)	0.08 (+0.08, 1.000)	0.0	13.2
	+F2	0.943 (+0.035, 0.453)	0.00 (−1.00, 0.002)	0.25 (+0.25, 0.250)	0.0	18.0
	F1+F2	0.956 (+0.048, 0.109)	0.00 (−1.00, 0.002)	0.00 (+0.00, 1.000)	0.0	18.0
gate	baseline	0.905	1.00	0.00	23.3	13.4
	+F1	0.897 (−0.008, 0.625)	1.08 (+0.08, 1.000)	0.00 (+0.00, 1.000)	21.9	13.3
	+F2	0.953 (+0.048, 0.146)	0.33 (−0.67, 0.021)	0.08 (+0.08, 1.000)	24.2	18.2
	F1+F2	0.982 (+0.077, 0.021)	0.00 (−1.00, 0.002)	0.00 (+0.00, 1.000)	24.9	18.1

TABLE XVI
SECOND EXTRACTOR `GEMINI-3.1-FLASH-LITE`: BASELINE VERSUS F1+F2, 12 PAIRED PERSONAS.

policy	variant	F1	stale	false	prompts/100	prompt tokens ($\times 10^3$)
auto	baseline	0.840	0.92	0.00	0.0	13.2
	F1+F2	0.892 (+0.053, 0.012)	0.00 (−0.92, <0.001)	0.00 (+0.00, 1.000)	0.0	18.1
gate	baseline	0.845	1.08	0.00	24.5	13.3
	F1+F2	0.901 (+0.057, 0.002)	0.00 (−1.08, 0.002)	0.00 (+0.00, 1.000)	26.0	18.4

TABLE XVII
PER-PERSONA RESULTS, FIRST MODEL, GATE POLICY. MISSED TARGETS: CHANGE TURNS WHOSE TARGET MEMORY WAS NOT IN THE PROMPT.

	baseline			F1+F2		
persona	F1	stale	missed targets	F1	stale	missed targets
0	1.000	0	0	1.000	0	0
1	0.857	2	1	1.000	0	0
2	0.933	1	1	1.000	0	0
3	0.933	0	0	0.875	0	0
4	0.909	1	2	1.000	0	0
5	0.857	1	2	1.000	0	0
6	0.824	2	1	1.000	0	0
7	0.857	1	1	1.000	0	0
8	0.923	1	1	1.000	0	0
9	0.909	1	1	0.909	0	0
10	0.933	1	1	1.000	0	0
11	0.923	1	2	1.000	0	0
mean	0.905	1.00	1.08	0.982	0.00	0.00

differ; in all three the automatic scorer labelled a correct statement of unemployment or finished project as “ended” or “other” while the annotator judged it true, so the automatic F1 is, if anything, slightly conservative on these three. These labels were kept as annotated. Since the scorer and the labels share one author and the same persona rules, the agreement shows consistency, not independence.

TABLE XVIII
PERSONA 1, BASELINE: PROPOSALS UNDER THE GATE.

session	action	proposal	accepted
3	update	The user goes to the gym at 7 am every day. → The user goes to the gym at 6 pm every day.	yes
3	update	The user is leading a project called Lotus at work. → The user is currently between projects at work as of October 2026.	no
4	forget	The user's birthday is on 21 September. → (removed)	yes
4	update	The user lives in Kolkata. → The user lives in Pune.	yes
5	update	The user lives in Pune. → The user lives in Chennai.	yes

TABLE XIX
PERSONA 1, BASELINE: FINAL PERSONA MEMORIES.

status	scored as	memory text
superseded	stale	The user goes to the gym at 7 am every day.
active	true	The user prefers their coffee as a cappuccino.
active	true	The user's dog is named Mithu.
superseded	stale	The user lives in Kolkata.
active	stale	The user drives a Seltos.
active	true	The user works at Swiggy.
active	stale	The user is leading a project called Lotus at work.
active	true	The user goes to the gym at 6 pm every day.
superseded	stale	The user lives in Pune.
active	true	The user bought a Swift car.
active	true	The user lives in Chennai.

X. EXTERNAL CHECK: LONGMEMEVAL

LongMemEval's knowledge-update questions ask for the latest value of a fact that changed during the history [4]. We took 30 such questions of the "S" setting and played each with all evidence sessions plus three random distractor sessions (the same for every system) through AURA's extractor, store and search; answers come from the ten memories that AURA's search returns, and are graded by a Gemini judge written after the benchmark's rule (the official prompt is not reproduced) and by string match. All runs use `gemini-3.1-flash-lite`. There is no label for which memory changes are right, so the gate's reviewer is an oracle approximation: it accepts proposals made while reading an evidence session and rejects the rest. This gate figure is therefore an upper-bracket, not a measurement of real users. Baselines are answering with no memory and with the raw sessions as context.

Table XXIV and Fig. 8 show that the gate system improves from 0.267 to 0.433 (0 questions worse, 5 better, $p = 0.063$), which is in the expected direction but not significant at this sample size; the systems without memory or with full context, which cannot be affected, differ by -0.033 between the two runs, which indicates run-to-run noise of that size. Auto-apply and append are not distinguishable between the runs. Full-context answering (0.733 in the second run) remains clearly above the gate system (0.433; 9 vs. 0 discordant questions, $p = 0.004$): a memory system trades accuracy for the properties it provides, such as small prompts, forgetting and user control. A run on all 78 knowledge-update questions would be needed for a firm statement.

TABLE XX
PERSONA 1, F1+F2: PROPOSALS UNDER THE GATE.

session	action	proposal	accepted
3	update	The user goes to the gym at 7 am every day. → The user goes to the gym at 6 pm every day.	yes
3	update	The user is leading a project called Lotus at work. → The user finished leading the project called Lotus and is currently between projects as of October 2026.	yes
4	forget	The user's birthday is on 21 September. → (removed)	yes
4	update	The user lives in Kolkata. → The user lives in Pune.	yes
5	update	The user lives in Pune. → The user lives in Chennai.	yes
5	update	The user drives a Seltos. → The user drives a Swift.	yes

TABLE XXI
PERSONA 1, F1+F2: FINAL PERSONA MEMORIES.

status	scored as	memory text
superseded	stale	The user goes to the gym at 7 am every day.
active	true	The user drinks their coffee as a cappuccino.
active	true	The user's dog is named Mithu.
superseded	stale	The user lives in Kolkata.
superseded	stale	The user drives a Seltos.
active	true	The user works at Swiggy.
superseded	stale	The user is leading a project called Lotus at work.
active	true	The user goes to the gym at 6 pm every day.
active	ended	The user finished leading the project called Lotus and is currently between projects as of October 2026.
superseded	stale	The user lives in Pune.
active	true	The user lives in Chennai.
active	true	The user drives a Swift.

XI. DISCUSSION

Where the guarantee ends. The gate provides a guarantee about what is *written*: nothing changes without approval. It provides none about what is *proposed*. In our experiments the second quantity depends on a component that looks like plumbing, the choice of candidate memories, and with default settings it is the main reason that stale facts remain. Evaluating a user-controlled memory by its write accuracy or by the number of approvals alone would miss this.

Design implications. (i) Report reachability alongside accuracy for any extractor-based memory. (ii) Search per sentence rather than per message when messages are mixed. (iii) Treat the minimum score as a trade-off: the unfiltered selection raises reachability (0.911 at $N = 350$) but admits unrelated memories, so the price of F2 is a larger prompt. (iv) Budget for more prompt tokens (about 35%) and slightly more review cards.

What the gate is for. The gate is a user-control mechanism as much as a safety mechanism. Every change to what the assistant believes about a person passes a visible decision, old values remain as history, and forgetting is an explicit operation that removes the memory. Our results show what this buys (protection against wrong writes) and what it costs (20–33 prompts per 100 turns in simulation, a little more with F2). They also show a limit: the gate's protection is only as good as

TABLE XXII
COST PER PERSONA (GATE POLICY; MEANS OVER 12 PERSONAS).

model	variant	requests	prompt tokens	output tokens	latency / request (s)
`gemini-3.5-flash-lite`	baseline	23.1	13 376	833	5.61
	F1+F2	23.1	18 072	796	9.39
`gemini-3.1-flash-lite`	baseline	23.1	13 347	932	6.15
	F1+F2	23.1	18 394	939	5.74

TABLE XXIII
AGREEMENT BETWEEN AUTOMATIC AND HAND LABELS (SINGLE ANNOTATOR).

run	labelled memories	agreement	rate	Cohen's κ
baseline	195	193	0.990	0.975
F1+F2	175	174	0.994	0.981

TABLE XXIV
LONGMEMEVAL KNOWLEDGE-UPDATE QUESTIONS (JUDGE-GRADED ACCURACY, $n = 30$). b/c: QUESTIONS CORRECT ONLY IN THE BASELINE / ONLY IN THE F1+F2 RUN.

system	baseline	F1+F2	difference	b/c	p
no memory	0.000	0.000	+0.000	0/0	1.000
full context	0.767	0.733	−0.033	1/0	1.000
append-only	0.400	0.333	−0.067	3/1	0.625
auto-apply	0.467	0.500	+0.033	1/2	1.000
gate (oracle reviewer)	0.267	0.433	+0.167	0/5	0.063

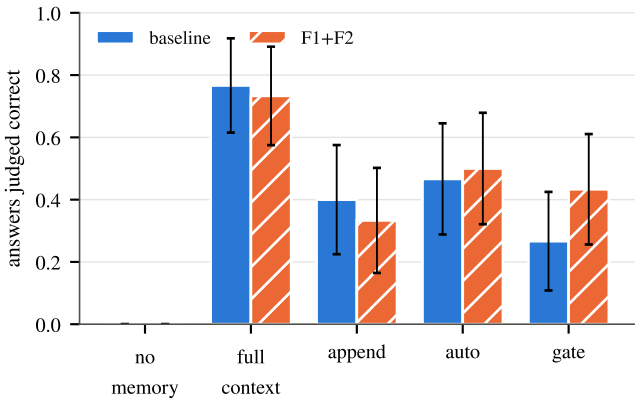


Fig. 8. LongMemEval accuracy by system for baseline and F1+F2 selection (95% normal-approximation bars).

the user's attention (Table VI), and adding prompts can lower that attention. Whether F2's additional prompts, which are mostly correct updates, are better tolerated than the baseline's is an empirical question for a user study.

Why the missed-update problem is easy to overlook. A missed update does not produce an error message, a wrong answer on the turn it occurs, or a prompt. It surfaces later, when the assistant states an outdated fact. Evaluations that look at single-turn extraction, or at accuracy of what is written, will not see it; measuring the final memory state against the true state, as we do, does.

Choosing between larger prompts and better search. F2 enlarges the prompt by about a third. Alternatives are a better retriever (for example a cross-encoder or a learned relevance gate over candidates, as in [8] for retrieval), a second extraction pass over the *whole* store for messages with a change cue, or a structured memory with typed slots in which an update names its slot. We did not evaluate these; the point of F2 is that a small, inexpensive change already removes most misses in our setting, and that reachability is a quantity worth tracking whichever remedy is chosen.

On F1. Letting a rejected change return only after new user input is cheap and conceptually neat, but we found no benefit with real models (the oracle user rejected no correct change in these runs), and only small, uncorrected-significance effects in simulation. Its value would have to be shown with real users who slip.

XII. THREATS TO VALIDITY

Synthetic and English personas. Personas, distractors and the wording of changes were written by the system's author. The change cues of F2 were designed after the reachability failures were seen on the same generator, so F2 may be tuned to it; the LongMemEval check is the only independent text and is small. **Small real samples.** 12 personas, one seed per cell and two models of the same family; exact sign tests with 12 pairs cannot resolve small effects, and several F1 results are inconclusive rather than null. **Simulated user.** The real runs use a perfect reviewer, the simulations a random slip. Real users may ignore, batch or rubber-stamp prompts; whether additional prompts are tolerated is open. No user study was done. **Single annotator and scorer.** Labels come from the author. **One embedding model.** Reachability depends on MiniLM and the 0.25 threshold. **LongMemEval.** Thirty questions, a judge that is not the official prompt, an oracle reviewer, and a single seed. **Multiple tests.** Reported as counts; only the large effects are robust to correction. **Related work.** Sources were read at abstract level.

XIII. FUTURE WORK

User study. The most important missing piece is a study with real users that measures how often they approve, reject or ignore proposals, whether they slip, and whether the extra prompts from F2 are tolerated. **Better candidate selection.** The city case (Table XIII) shows that sentence search with a larger candidate set is not enough when a statement contains no word that points to the old value; slot-typed memories or a second pass are candidates. **Other languages and models.** All

personas are English and both extractors are Gemini models. **Bigger real samples.** More personas, more seeds, and all 78 LongMemEval knowledge-update questions would turn several inconclusive results into findings. **Proactive use of memory.** AURA also contains an optional layer that asks the user whether a stored fact is still true, sends reminders and prepares briefings; its checks use the same proposal path as the gate. We did not evaluate it and make no claim about it here.

XIV. Conclusion

A review gate on memory changes is effective against the errors it can see: in our experiments it removes almost all of the true facts that automatic application destroys. It does not address updates the extractor never proposes, and in a retrieval-based design these are governed by which memories the extractor is shown. Showing candidates by sentence, without a minimum score and with a larger set on a change cue, removed all observed misses for two real Gemini models and removed the stale facts, at roughly a third more prompt tokens and no significant increase in review prompts, while an external check on LongMemEval pointed in the same direction without reaching significance. The practical lesson for user-controlled memories is to measure and report what the extractor can reach. Evaluating with real users, in other languages and with larger real samples is the necessary next step; the evaluation code and every raw result file are provided so that others can repeat and extend it.

Data and Code Availability

The evaluation harness (`eval/`), the persona generator, all raw per-run files and the scripts that produce every table and figure of this paper (`eval/make_paper_assets.py`) are in the AURA project repository. Numbers in the text are generated from the result folders.

Appendix A
Scoring Rules and Labels

Value matching. Each persona slot has a set of possible values (for example the gym slot has "6 am", "7 am", "6 pm", "8 pm"), each with alternative spellings ("6am", "6:00 am", "18:00"); birthdays accept "12 March", "March 12", "12th March" and so on; coffee values accept "black coffee" or "oat milk". A value is matched as a whole word, case-insensitively. Distractor values (a hypothetical city, a friend's car) have their own matchers.

Wording of third parties, hypotheticals and ended facts. A memory that mentions only a distractor value is *third-party* if it contains a word for a relative, friend, colleague or neighbour, *hypothetical* if it contains "if", "would" or "hypothetical", and *false* otherwise. A memory about a retired fact is *ended* if it contains a phrase such as "no longer", "finished", "wrapped up", "between projects", "stopped", "sold", "unemployed" or "not working" and the fact was retired but not asked to be forgotten. A fact that the user asked to forget and that is still stored counts as *stale*.

Annotation protocol. For every memory in the baseline and F1+F2 real runs the annotator saw the persona's sessions, the true final state and the memory text, and wrote one of seven labels (true, ended, stale, false, hypothetical, third-party, other), looking at the automatic label only afterwards where possible. The automatic and the hand label were then compared (Table XXIII).

Appendix B
Complete Policy Results

Table XXV gives all metrics of the policy comparison for all error rates.

TABLE XXV
All policies, all extractor error rates (simulated, $u = 0.05$; mean per persona).

e	policy	F1	precision	recall	stale	false	destroyed	wrong applied	prompts/100
0	append-only	0.597	0.483	0.791	5.00	0.00	0.00	0.00	0.0
	auto-apply	0.881	0.798	0.992	1.57	0.00	0.00	0.00	0.0
	gate changes	0.854	0.768	0.968	1.80	0.00	0.00	0.00	19.5
	gate all	0.861	0.788	0.959	1.62	0.00	0.00	0.00	65.8
0.05	append-only	0.584	0.474	0.769	4.90	0.12	0.00	0.00	0.0
	auto-apply	0.840	0.765	0.944	1.65	0.17	0.78	0.82	0.0
	gate changes	0.820	0.741	0.929	1.90	0.12	0.05	0.07	22.2
	gate all	0.818	0.753	0.908	1.82	0.00	0.03	0.05	67.3
0.1	append-only	0.581	0.478	0.753	4.58	0.33	0.00	0.00	0.0
	auto-apply	0.798	0.730	0.898	1.68	0.43	1.60	1.75	0.0
	gate changes	0.797	0.725	0.903	1.85	0.33	0.03	0.05	24.4
	gate all	0.795	0.749	0.867	1.83	0.02	0.05	0.05	69.0
0.2	append-only	0.549	0.457	0.703	4.25	0.70	0.00	0.00	0.0
	auto-apply	0.735	0.675	0.831	1.73	0.83	2.83	3.10	0.0
	gate changes	0.733	0.666	0.836	1.98	0.70	0.18	0.23	28.8
	gate all	0.765	0.732	0.818	1.88	0.00	0.13	0.17	72.5
0.3	append-only	0.523	0.440	0.665	3.87	1.20	0.00	0.00	0.0
	auto-apply	0.650	0.596	0.743	1.92	1.33	4.60	4.97	0.0
	gate changes	0.653	0.585	0.765	2.18	1.22	0.25	0.28	33.4
	gate all	0.707	0.700	0.743	1.98	0.07	0.40	0.43	77.6

Appendix C
Complete Ablation Results

Tables XXVI and XXVII give every cell of the simulated ablation. An asterisk marks a sign-test $p < 0.05$ against the baseline (uncorrected; 210 tests in all). The upper three blocks vary the extractor error rate ($u = 0.05$), the lower four the user slip rate ($e = 0.1$; gate only).

Appendix D
Reachability for All Store Sizes

Table XXVIII gives reachability and mean candidate count for all selection modes, store sizes and turn types.

Appendix E
Reproducing the Results

All commands are run from the repository root. Simulated experiments need no API key; real-model experiments need `GEMINI_API_KEY` and internet access.

- `npm run eval:gate`, `eval:retrieval`, `eval:reach`: policies, retrieval, reachability;
- `npm run eval:gate -- --errors 0,0.1,0.2 --policies auto,gate --restate 1 --candidates adaptive --repropose 2 --out eval/results/ablation/both` (and the three other variants), then `npm run eval:compare` for the paired statistics;
- `npm run eval:real -- --yes`: real-model runs (twelve personas, four variants, one folder each); `npm run eval:rescore`: scoring and annotation sheet;

TABLE XXVI
ABLATION, GATE POLICY: ALL CELLS (60 PAIRED PERSONA RUNS EACH). * : UNCORRECTED $p < 0.05$.

e	u	variant	F1	stale	false	exact state	prompts/100
0	0.05	baseline	0.854	1.80	0.00	0.033	17.6
		+F1	0.865*	1.70*	0.00	0.050	18.3*
		+F2	0.950*	0.40*	0.00	0.667*	24.3*
		F1+F2	0.967*	0.27*	0.00	0.750*	25.0*
0.1	0.05	baseline	0.817	1.83	0.33	0.067	23.7
		+F1	0.821	1.80	0.33	0.067	23.9
		+F2	0.890*	0.65*	0.33	0.267*	29.1*
		F1+F2	0.901*	0.58*	0.33	0.300*	29.4*
0.2	0.05	baseline	0.737	1.98	0.80	0.017	28.6
		+F1	0.747	1.92	0.80	0.017	28.8
		+F2	0.776*	1.22*	0.82	0.083	32.2*
		F1+F2	0.784*	1.17*	0.82	0.083	32.7*
0.1	0	baseline	0.826	1.75	0.33	0.083	23.7
		+F1	0.826	1.75	0.33	0.083	23.8
		+F2	0.900*	0.57*	0.33	0.350*	29.1*
		F1+F2	0.903*	0.55*	0.33	0.367*	29.2*
0.1	0.1	baseline	0.801	1.98	0.35	0.050	23.6
		+F1	0.809	1.92	0.35	0.050	24.2*
		+F2	0.861*	0.97*	0.33	0.233*	28.9*
		F1+F2	0.881*	0.82*	0.33	0.267*	29.9*
0.1	0.2	baseline	0.771	2.32	0.35	0.000	23.3
		+F1	0.793*	2.20*	0.33	0.000	24.5*
		+F2	0.829*	1.25*	0.37	0.050	28.6*
		F1+F2	0.851*	1.10*	0.37	0.050	30.1*
0.1	0.3	baseline	0.740	2.60	0.38	0.017	23.3
		+F1	0.750	2.53	0.38	0.017	25.3*
		+F2	0.765*	1.80*	0.35	0.033	28.6*
		F1+F2	0.800*	1.57*	0.33	0.033	31.0*

TABLE XXVII
ABLATION, AUTO-APPLY POLICY: ERROR-RATE CELLS (60 PAIRED PERSONA RUNS EACH). * : UNCORRECTED $p < 0.05$.

e	u	variant	F1	stale	false	exact state	prompts/100
0	0.05	baseline	0.881	1.57	0.00	0.083	0.0
		+F1	0.881	1.57	0.00	0.083	0.0
		+F2	0.989*	0.08*	0.00	0.917*	0.0
		F1+F2	0.989*	0.08*	0.00	0.917*	0.0
0.1	0.05	baseline	0.819	1.62	0.40	0.000	0.0
		+F1	0.819	1.62	0.40	0.000	0.0
		+F2	0.899*	0.43*	0.42	0.050	0.0
		F1+F2	0.899*	0.43*	0.42	0.050	0.0
0.2	0.05	baseline	0.750	1.62	0.93	0.000	0.0
		+F1	0.750	1.62	0.93	0.000	0.0
		+F2	0.800*	0.88*	0.87	0.000	0.0
		F1+F2	0.800*	0.88*	0.87	0.000	0.0

TABLE XXVIII
REACHABILITY (MEAN NUMBER OF CANDIDATES) BY MODE, STORE SIZE N AND TURN TYPE.

N	turn type	default	no min.	sentence	adaptive
0	single sentence	0.898 (8.2)	1.000 (11.4)	0.898 (8.2)	1.000 (11.4)
	mixed turn	0.747 (8.1)	0.998 (11.4)	0.898 (8.2)	1.000 (11.9)
25	single sentence	0.752 (8.8)	0.990 (17.4)	0.752 (8.8)	0.993 (23.8)
	mixed turn	0.439 (8.2)	0.903 (17.4)	0.752 (8.8)	0.980 (24.2)
50	single sentence	0.749 (9.1)	0.975 (18.5)	0.749 (9.1)	0.986 (25.9)
	mixed turn	0.364 (8.3)	0.822 (18.5)	0.749 (9.1)	0.969 (25.9)
100	single sentence	0.727 (9.6)	0.952 (19.2)	0.727 (9.6)	0.969 (27.1)
	mixed turn	0.344 (8.3)	0.745 (19.3)	0.727 (9.6)	0.921 (27.2)
200	single sentence	0.704 (10.4)	0.934 (19.6)	0.704 (10.4)	0.951 (27.8)
	mixed turn	0.305 (8.3)	0.668 (19.5)	0.704 (10.4)	0.900 (27.8)
350	single sentence	0.690 (11.3)	0.911 (19.8)	0.690 (11.3)	0.936 (28.2)
	mixed turn	0.286 (8.3)	0.607 (19.8)	0.690 (11.3)	0.866 (28.1)

- `npm run eval:lme -- --yes`: LongMemEval subset; `npm run eval:compare-lme`;
- `python eval/make_paper_assets.py --root eval --out paper`: every table, number macro and the real-run and LongMemEval figures of this paper.

REFERENCES

[1] C. Lam, J. Li, L. Zhang, and K. Zhao, "Governing evolving memory in LLM agents: Risks, mechanisms, and the stability and safety governed memory (SSGM) framework," *arXiv preprint arXiv:2603.11768*, 2026.

[2] P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav, "Mem0: Building production-ready AI agents with scalable long-term memory," *arXiv preprint arXiv:2504.19413*, 2025.

[3] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, and Y. Fang, "Evaluating very long-term conversational memory of LLM agents," *arXiv preprint arXiv:2402.17753*, 2024.

[4] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu, "LongMemEval: Benchmarking chat assistants on long-term interactive memory," *arXiv preprint arXiv:2410.10813*, 2024.

[5] V. Patel, "Supersede: Diagnosing and training the memory-update gap in LLM agents," *arXiv preprint arXiv:2606.27472*, 2026.

[6] Z. Xiong, Y. Lin, W. Xie, P. He, Z. Liu, J. Tang, H. Lakkaraju, and Z. Xiang, "How memory management impacts LLM agents: An empirical study of experience-following behavior," *arXiv preprint arXiv:2505.16067*, 2025.

[7] S. Xiao, S. Wang, X. Chen, K. Chao, M. Cui, F. Qian, F. Meng, C. Mei, C. Jiang, Q. Ouyang, and J. Yi, "When errors become memories: Causal pathway tracing in multi-turn memory-augmented LLMs," *arXiv preprint arXiv:2608.30198*, 2026.

[8] J. Zhang, K. Chen, J. Ma, Y. Hu, L. He, Y. Zhang, J. Liu, X. Yang, T. Zhang, and R. Jia, "Beyond similarity: Trustworthy memory search for personal AI agents," *arXiv preprint arXiv:2606.06054*, 2026.

[9] K. Chu, "A systematic survey of security threats and defenses in LLM-based AI agents: A layered attack surface framework," *arXiv preprint arXiv:2604.23338*, 2026.

[10] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," *arXiv preprint arXiv:2002.10957*, 2020.

[11] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.

[12] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.